

Strangers poked my kiosk for five days, and it never went down.

I start with **the person who is stuck** and what they are trying to get done, then take a vague brief all the way to **a working version people can use**, and read the data to decide what changes next. The part I care about most is **packaging**, making something clear and good-looking enough that people want to keep using it.

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Booth Host Kiosk

SHIPPED · FINANCIAL COMPANY

The kiosk ran all 5 days of the event with zero downtime, and I shipped same-day fixes while it was live.

5 days live with zero downtime

02

VivaTech Conversation Analysis

ANALYSIS DONE · KLEON, FOR THE SAMSUNG BOOTH

A report that proposed a goal mode, an A/B test to prove it, and two ways to earn from the avatar's attention.

2 ways to earn from the avatar's attention

03

Glow Buddy

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A working prototype in 3 weeks, packaged as an Android app for foldable phones and refined over 99 builds with feedback from the client's agency and visitors.

3 wks to a working prototype

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In progress. The goal is a ranked list of design changes for each user type and cluster.

6 user types in the study

05

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The concept behind a Phase 1 research and prototype program with a university medical research lab.

3 supporter types in the concept

01

SHIPPED

Booth Host Kiosk

A voice and touch kiosk that I designed, built, and ran live for a financial company's booth at a large international event.

ROLE

Sole designer and builder

CLIENT

Financial company

TIMELINE

2026, live for 5 days

PLATFORM

9:16 touch kiosk, Korean and English

5 days

live with zero downtime

52

voice clips

8+

screens, voice and touch

AT A GLANCE**MY ROLE**

Sole designer and builder, and the person who kept it running live on the floor.

PROBLEM

A financial company's booth had to greet and hold a large international crowd, and nobody else was assigned to design or build it.

OUTCOME

The kiosk ran all 5 days of the event with zero downtime, and I shipped same-day fixes while it was live.

TARGET

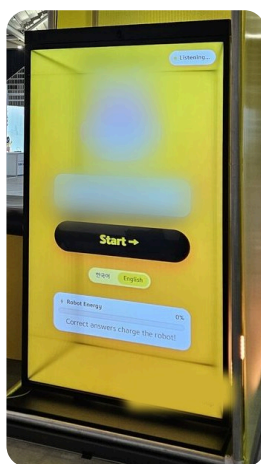
Visitors walking past a busy booth who speak different languages and have never used a kiosk like this before

IMPACT

I shipped same-day fixes while the kiosk was running and checked each one on the live screen within about 60 seconds

LEADS TO

Proof that Klleon can ship and run a live interactive install for an event client



The kiosk on the exhibition floor, with the client's branding blurred

THE FULL STORY**01 Context**

Klleon's founder handed me this project with a single line, which was to make a small robot that people could talk to. A financial company needed its booth to greet a large international crowd, explain what was in the exhibition, and keep

people around long enough to care about the brand. There was no designer and no engineer assigned to it, so every screen and every line of code started with me.

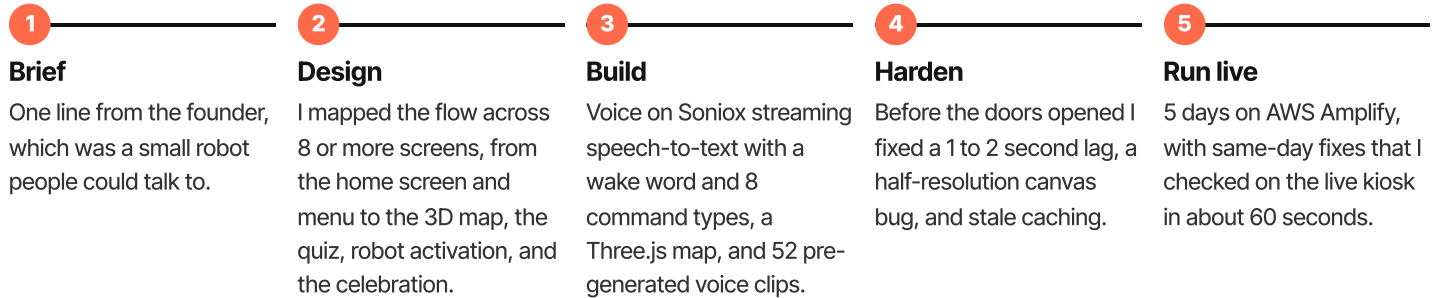
02 Users and insight

I pictured someone walking past the booth with a few seconds of attention, who might not speak Korean, and who has no reason to trust a screen they have never seen before. Because of that, I decided early that the kiosk should start the conversation itself, so that nobody would have to figure out where to tap first.

03 Goals and what success meant

- ✓ A visitor can go from walking past the booth to talking with the mascot without any help from staff.
- ✓ The kiosk stays up for all 5 days.
- ✓ Narration is clear in both Korean and English on a loud exhibition floor.
- ✓ People have a reason to stay after the greeting, which is the job of the quiz.

04 Process



05 What I built

- 01 Voice greeting and commands**
A mascot greets visitors and answers to her name. I built the voice side on Soniox streaming speech-to-text with a wake word and 8 voice command types, such as start, map, and quiz, each with many phrasings in both languages, and I generated 52 voice clips, 26 prompts in Korean and English, so every prompt had both versions.
- 02 3D map of the hall**
I built a 3D model of the exhibition hall in Three.js with about 13 booths on it, and each booth has its own spoken narration, so the kiosk was useful to people who just wanted to find something.
- 03 True or false quiz**
I added a two-round true or false quiz where every right answer charges up a robot, which ends in a celebration screen, because I wanted people to have a reason to stay at the booth after the greeting.
- 04 Louder English narration**
The exhibition floor was loud, so I used the Web Audio API to push the English narration to 1.5 times the browser's normal volume limit, while keeping the Korean audio exactly as it was.

06 Key decisions and tradeoffs

I CHOSE

Voice first, with touch as the backup

VS

INSTEAD-OF

A touch menu like most kiosks use

BECAUSE Touch alone is cheaper and never mishears, but a visitor who does not read Korean can start just by saying the mascot's name. The cost was speech recognition on a loud floor, so I added a wake word, many phrasings per command in both languages, and cooldowns against false triggers, and kept touch as the backup.

I CHOSE

Reliability over extra features

VS

INSTEAD-OF

Adding more screens and effects before the event

BECAUSE A crash in front of an international crowd could not be undone, so I spent the last stretch before the event removing lag and fixing caching, because a new effect was worth less than a kiosk that stayed up.

07 Engineering for a live event

Before the event, I traced a 1 to 2 second lag to a drop shadow redrawn every frame and an animation loop that never stopped, fixed a half-resolution canvas that tinted half the screen, cut the main images by about 28 to 40 percent, and set up cache headers and versioned files so every fix reached the kiosk. During the 5 days I shipped same-day fixes through AWS Amplify, checked each one on the live kiosk within about 60 seconds, and kept the speech keys on the server as single-use keys that expire after 300 seconds.

08 Results

The kiosk ran for all 5 days of the event with zero downtime.

09 What I learned

LESSON 1

When a correct fix does not seem to work, I now check for stale files before I rewrite any code, because caching caused most of my scariest moments.

LESSON 2

I shipped without usage tracking, so I can prove the kiosk stayed up but not how many people finished the quiz. Now I decide what to measure and start logging it before launch.

LESSON 3

During a live event I could not fully test ahead of time, I only shipped fixes I was sure could not break a screen that already worked.

10 What's next

Kileon can now show event and retail clients an install that ran in public for 5 days. On the next install I would add session and quiz-completion tracking from day one, so the report covers how many people finished the quiz as well as the uptime.

WHERE THIS SITS IN THE PRODUCT LOOP

Discover

Define

Design

Build

Launch

Measure

Iterate

SKILLS AND TOOLS

Voice UX

Live ops

Performance

Three.js

Next.js

Web Audio API

Soniox streaming STT

AWS Amplify

02

ANALYSIS DONE

VivaTech Conversation Analysis

I analyzed the conversation logs from a Kileon avatar that ran the front desk of the Samsung booth at VivaTech 2026 in Paris, found that small talk almost never turned toward a product on its own, and designed the A/B test to change that.

ROLE

Analysis and report, on my own

CLIENT

Kileon, for the Samsung booth

TIMELINE

June 2026, VivaTech in Paris

PLATFORM

Conversation logs in several languages

2

ways to earn from the avatar's attention

1

A/B test designed for the next event

AT A GLANCE

MY ROLE

I pulled the logs, cleaned them, ran every number, and wrote the report on my own.



PROBLEM

Kileon knew visitors talked to its avatar at the Samsung booth, but not what they talked about or how that attention could turn into something a client pays for.



OUTCOME

A report that proposed a goal mode, an A/B test to prove it, and two ways to earn from the avatar's attention.

TARGET

Brands like Samsung that put a Kileon avatar on their floor and want visitors to reach a product, a signup, or a demo

IMPACT

Showed that visitors gave the avatar sustained attention but small talk rarely turned toward a product, and turned that into two revenue paths and a test plan

LEADS TO

A goal mode Kileon can build once and switch on for any brand partner



Kileon's avatar working the front desk of the Samsung booth at VivaTech 2026

THE FULL STORY

01 Context

Klleon ran one of its avatars as the AI promoter at the front desk of the Samsung booth at VivaTech 2026 in Paris, answering visitors in several languages. After the event I pulled the conversation logs to find out what visitors did with it and where the business value was.

02 Cleaning the data first

A session-timeout bug had merged many visitors into a few oversized sessions for the first few days. I cut those days and rebuilt every number on the clean days only, because merged sessions would have made every per-session rate wrong.

03 What I found

LESSON 1

Visitors stayed for long conversations, and most of them came back with a second question.

LESSON 2

Most of what visitors said went to the avatar's personality, even though only a small slice of the knowledge base was built for it.

LESSON 3

Left alone, a conversation almost never moved from small talk to a product.

LESSON 4

Conversations that opened on a Samsung topic reached a product far more often than ones that stayed in small talk.

04 Key decisions and tradeoffs

I CHOSE

Cut the buggy days entirely

VS

INSTEAD OF

Keeping them and adjusting for the bug

BECAUSE I could not trust any per-session rate built on merged sessions.

I CHOSE

Propose a goal mode where the avatar steers a turn or two in

VS

INSTEAD OF

Loading more product facts into the knowledge base

BECAUSE Visitors already talked past most of the facts, and the way a conversation opened already predicted where it ended up.

05 The test I designed

At the next event, run the current passive avatar against one that steers toward what the partner cares about, such as a product, a signup, or a demo booking. The main metric is the share of visitors who reach that outcome, measured against the baseline from this event.

06 Two ways to earn from it

LESSON 1

Intent data. Every conversation records what visitors asked, in which language, and where they wanted to go, which a brand can use on its own.

LESSON 2

Goal mode. A feature built once and switched on for any brand partner, sold as an outcome a client can put a number on.

07 What I learned

LESSON 1

Before I compute anything from an export, I now check that each session is a single visit, because the timeout bug would have inflated several days of results if I had taken the export as it was.

08 Status

The report proposed the test for the next event. I have not put a dollar figure on goal mode yet, because the lead and token costs were not in this dataset. The exact numbers stay internal to Klleon.

WHERE THIS SITS IN THE PRODUCT LOOP

Discover › Define › Design › Build › Launch › Measure › Iterate

SKILLS ● AND TOOLS ●

Data analysis Experiment design Data cleaning Funnel analysis A/B test design Monetization

03

PROTOTYPE

Glow Buddy

A digital human for a phone maker's stores that either answers spoken questions about the phone a customer is holding or takes them on a passport mission built around the way a foldable phone opens.

ROLE

Concept, design and build

CLIENT

Phone maker, retail

TIMELINE

Prototype in 3 weeks, 99 builds

PLATFORM

Foldable phone, Android app

3 wks

to a working prototype

99

builds

AT A GLANCE

MY ROLE

Concept, design, and build, on my own from the first idea to the working app.



PROBLEM

Shoppers who pick up a new foldable in a store often have nobody to explain it, and Kileon needed a use for its avatar that the phone maker would want too.



OUTCOME

A working prototype in 3 weeks, packaged as an Android app for foldable phones and refined over 99 builds with feedback from the client's agency and visitors.

TARGET

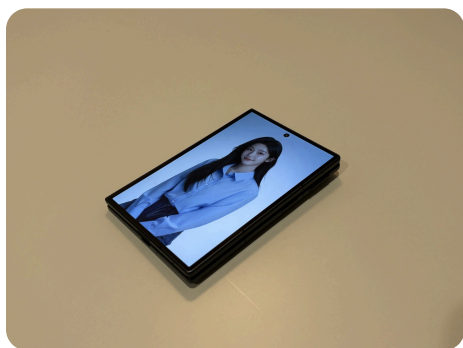
Shoppers who pick up a new phone in a store and have nobody nearby to explain it to them

IMPACT

A working prototype in 3 weeks, refined over 99 builds of testing and fixes

LEADS TO

A use case the phone maker can test on its display phones



The avatar on a foldable phone



A visitor's finished book cover



Me with one of Kileon's avatars during testing

THE FULL STORY

01 Context

Klleon's founder wanted our digital humans running on every phone on display in a major phone maker's stores, and he asked me to find the reason the phone maker would want that. My answer was that a customer who picks up a phone in a store should not have to figure it out alone. His only other direction was to make it a mission, plus a mention that one of the brand's partners could be interesting, so every feature that made it in, and every feature I cut, was my call.

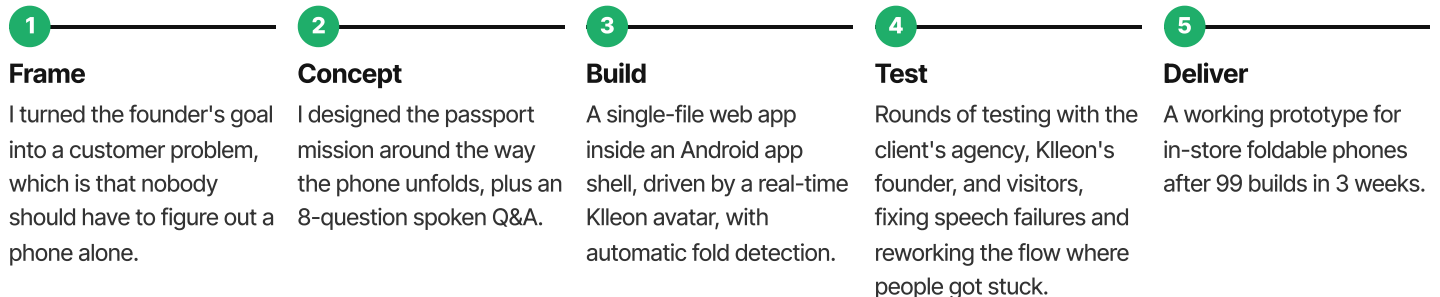
02 Users and insight

I saw two kinds of shoppers, which I started calling explorers and deciders. Explorers want an experience and are happy to play, and deciders just want a quick answer about the phone before they buy. So the app had to give deciders a fast way to ask their question without sitting through the mission first. A foldable phone opens like a book, so I made opening it part of the story, and a visitor builds their own passport while using the one feature that makes a foldable different from other phones.

03 Goals and what success meant

- ✓ A shopper can get an answer about the phone just by asking out loud.
- ✓ Explorers leave with something to keep, which is the certificate with a QR code.
- ✓ Opening the phone becomes part of the story, so using the app means using the feature that sets the phone apart.
- ✓ The app can be rebuilt for other phone models without starting over.

04 Process



05 What I built

- 01 The passport mission**
The visitor takes a selfie, picks a city, watches a short travel scene, takes an arrival photo with a filter for that city, and gets a certificate with a QR code they can download and take home.
- 02 Fold detection**
The narrow cover screen and the wide inner screen each have their own layout. I wrote fold detection that reads the screen's aspect ratio, with a buffer so it does not flicker, and switches layouts on its own the moment the phone opens.
- 03 Spoken Q&A**
For people who only want to know about the phone, the avatar answers an 8-question spoken Q&A about the device in their hand.
- 04 Custom book cover**
Visitors design their own book cover, which ties back to one of the brand's partnerships.

06 Key decisions and tradeoffs

I CHOSE

Two paths, a mission and a quick spoken Q&A

VS

INSTEAD OF

One mission that everyone has to go through

BECAUSE Some shoppers only want an answer before they decide, and making them play a game first would lose them.

I CHOSE

Automatic fold detection that switches layouts on its own

VS

INSTEAD OF

A button asking the visitor to switch modes

BECAUSE The moment someone opens the phone is the moment the story should move forward, so the app reacts by itself and uses a buffer so it does not flicker.

07 Testing and iteration

I tested it with the client side and with visitors over many rounds. Along the way I fixed text-to-speech and speech-to-text failures, reworked the mission flow when people got stuck, and redesigned the interface more than once.

08 Results

The prototype was working in 3 weeks and went through 99 builds based on feedback from the client's agency, Kileon's founder, and visitors.

09 What I learned

LESSON 1

Speech features that work fine at a desk can fail in a real store, so I now test voice with real people as early as I can.

LESSON 2

Because the founder left the scope to me, most of my time went into deciding what to build, and I learned to judge every feature by whether it helped the explorer, the decider, or neither.

10 What's next

Next I would push to rebuild it for more phone models and measure how many shoppers finish the mission compared with how many only ask questions, because that split would tell the client which path to put first.

WHERE THIS SITS IN THE PRODUCT LOOP

Discover › Define › Design › Build › Launch › Measure › Iterate

SKILLS AND TOOLS

Interaction design Prototyping Scoping Interaction design Fold detection Scene state machine
Kileon avatar SDK Speech Q&A Android

04

IN PROGRESS

Lab Platform Navigation Study

A biotech R&D software company is introducing new navigation to its lab data platform, and our team is finding out how each type of user reacts to it and which design changes would make them want to switch.

ROLE	COMPANY	TIMELINE	FOCUS
Contract Product Design Analyst	Biotech R&D software company	Sep 2026 to present	Navigation

6

user types in the study

AT A GLANCE

MY ROLE

Contract Product Design Analyst on the research team.

PROBLEM

New navigation only pays off if the scientists and admins who use the platform every day adopt it.

OUTCOME

In progress. The goal is a ranked list of design changes for each user type and cluster.

TARGET

Scientists and admins who use a lab data platform every day, split into 6 user types

IMPACT

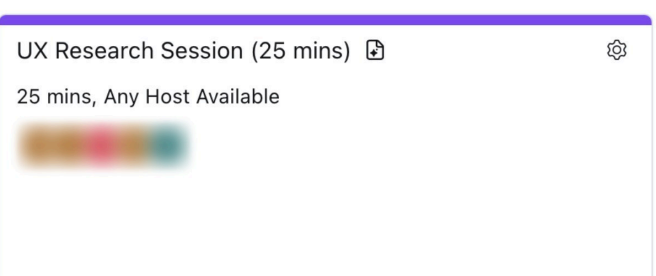
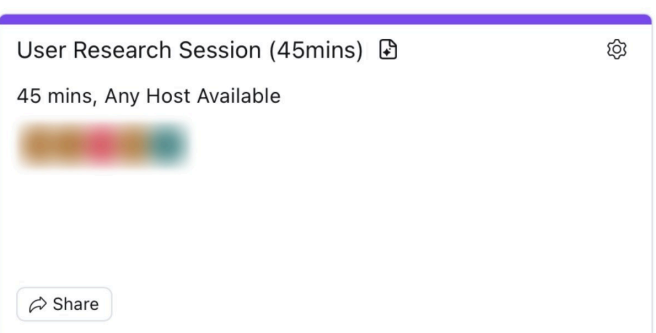
When the sessions finish, design changes ranked for each user type, so the team knows what to build first and for whom

LEADS TO

A navigation rollout that each group of users is more likely to adopt on their own



The kind of dense lab workspace the new navigation has to organize



How we set up the 1:1 research sessions, with names blurred

THE FULL STORY

01 Context

The client makes a lab data platform that biotech and pharma teams use every day, and there is a lot in it to find. The company is now introducing new navigation concepts, and it needs the people who use the platform every day to adopt them.

02 The research question

We want to understand how core users move around the platform today, how they react to the new navigation, how those reactions differ from one type of user to another, and what changes would make them eager to switch.

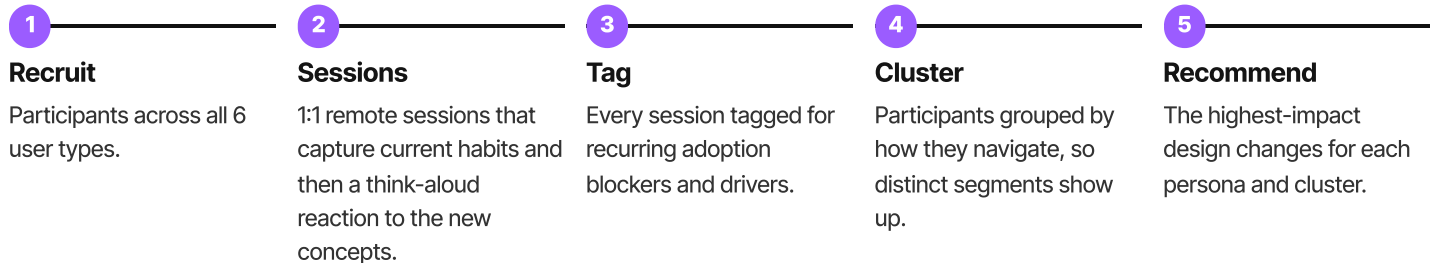
03 Users

We are recruiting across 6 user types, from admins to bench scientists, computational biologists, team leads, molecular biologists, and chemists. Each of them uses the platform for different work, so I expect each of them to look for things in different places.

04 Goals and what success means

- ✓ Understand how each of the 6 user types moves through the platform today.
- ✓ Capture first reactions to the new navigation through think-aloud sessions.
- ✓ Find the blockers and drivers of adoption that come up again and again.
- ✓ Rank design changes by impact for each persona and cluster, so the team knows what to build first.

05 Process



06 What a session looks like

Each session is a 1:1 remote call with one person from one of the 6 user types, and we booked two formats, a 45-minute user research session and a shorter 25-minute UX session. In the first part we look at how the person gets around the platform today for their normal work. In the second part we show the new navigation concepts and ask them to think out loud as they react, so we hear the reason behind each reaction along with the reaction itself.

07 How we will read the data

01 Tag every session

We tag each session for the blockers that stop someone from adopting the new navigation and the drivers that make them want it, so the same patterns show up across different people.

02 Compare by user type

We compare reactions across the 6 user types, because each type uses the platform for different work and looks for things in different places.

03

Cluster by behavior

We group participants by how they navigate, which can split one job title into several distinct segments.

04

Rank the changes

For each user type and each cluster, we rank the design changes by how much they would help people adopt the new navigation.

08 Key decisions and tradeoffs

I CHOSE

1:1 think-aloud sessions

VS

INSTEAD OF

A survey sent to many users

BECAUSE Navigation problems show up in what people do and say in the moment, which a survey answer usually misses.

I CHOSE

Clustering people by how they navigate

VS

INSTEAD OF

Only grouping people by job title

BECAUSE Two scientists with the same title can move through the product very differently, and behavior-based clusters show that difference.

09 Who I work with

 The client's product team

 Our research team

10 Deliverables

We will hand over a final presentation with case studies for each persona, along with an organized spreadsheet of all the session data, so the product team can keep slicing it after we are gone.

11 Status

The study runs this fall. Recruiting and the first sessions come first, then tagging and clustering, and then the ranked recommendations and the final presentation.

12 What's next

If we get this right, the client can roll out its new navigation in a way that matches how each group already works, which makes adoption much more likely to happen on its own. Once the sessions are done, the next step is turning the top-ranked changes into designs the team can build and test.

WHERE THIS SITS IN THE PRODUCT LOOP

Discover

>

Define

>

Design

>

Build

>

Launch

>

Measure

>

Iterate

SKILLS ● AND TOOLS ●

User research

Synthesis

Segmentation

Information architecture

User interviews

Think-aloud testing

Behavioral clustering

Persona segmentation

Information architecture

05

IN PROGRESS

Caregiver Support System

Kleon and a university medical research lab are developing a real-time AI companion for family caregivers, and I shaped the product concept and the pitch.

ROLE

Product concept and pitch

PARTNERS

Kleon x a university medical lab

STAGE

Phase 1, research and prototype

PLATFORM

Mobile, real-time video avatar

3

supporter types in the concept

30+

caregivers planned for Phase 1 co-design

AT A GLANCE

MY ROLE

Product concept and pitch.



PROBLEM

Unpaid family caregivers often have nobody to turn to when they need help late at night.



OUTCOME

The concept behind a Phase 1 research and prototype program with a university medical research lab.

TARGET

Family caregivers who look after a relative without pay

IMPACT

The concept and pitch behind a Phase 1 research and prototype program with an academic medical partner

LEADS TO

A clinical validation trial and a path to users through a major health app in later phases



Concept screen · choosing an AI supporter

THE FULL STORY

01 Context

About 59 million Americans care for an adult loved one, mostly without pay (AARP, 2026). Caregiver burnout is linked to worse outcomes for the person being cared for, and the support that exists today is scattered and generic, so nobody is there at 11pm when a caregiver needs help right away.

02 Users and insight

I kept picturing one person, which is a caregiver late at night who is tired, who has a practical question, and who also just needs someone to talk to. Most tools today handle the question or the feelings, but rarely both, so we wanted one conversation that could hold both.

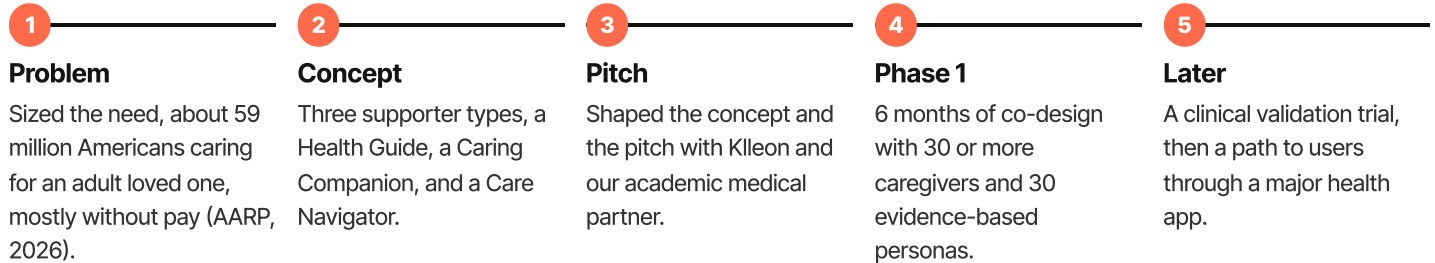
03 Goals and what success means

✓ Caregivers get practical and emotional support in the same conversation.

✓ People can pick the kind of support they need in the moment.

✓ The product is shaped by caregivers before it is built.

04 Process



05 The concept

01 Three supporter types

Caregivers choose the AI supporter that fits the moment, which could be a Health Guide for clear information, a Caring Companion for emotional support, or a Care Navigator for next steps.

02 Live video avatar

Each supporter is a Klleon digital human that talks back live, so it feels like talking to a person who is there with you.

03 Practical and emotional help in one chat

A caregiver can vent and get a concrete next step in the same conversation, without switching between apps.

06 Key decisions and tradeoffs

I CHOSE
Three supporter types to choose from

VS

INSTEAD-OF
One general assistant

BECAUSE A caregiver at 11pm might need facts, comfort, or a plan, and letting them choose keeps each conversation focused on that one need.

I CHOSE
A live video avatar

VS

INSTEAD-OF
A text chatbot

BECAUSE Seeing a face that talks back makes it feel like a person is there, which matters most when someone is exhausted.

I CHOSE

Co-designing with 30 or more caregivers first

VS

INSTEAD OF

Building a prototype first and testing it later

BECAUSE The personas should come from caregivers themselves, so the product fits their lives from the start.

07 Who I worked with

 Kleon

 Our academic medical partner

08 The plan

Phase 1 is a 6-month research and prototype program. We will co-design with 30 or more caregivers, build 30 evidence-based personas from that research, and deliver a working AI prototype that has been shaped by their feedback.

09 What I learned

LESSON 1

Because the partner was a medical research lab, I led the pitch with evidence, so the concept starts from the AARP numbers and ends with a clinical validation plan.

10 What's next

Later phases plan a clinical validation trial and then a path to users through a major health app.

WHERE THIS SITS IN THE PRODUCT LOOP

Discover › Define › Design › Build › Launch › Measure › Iterate

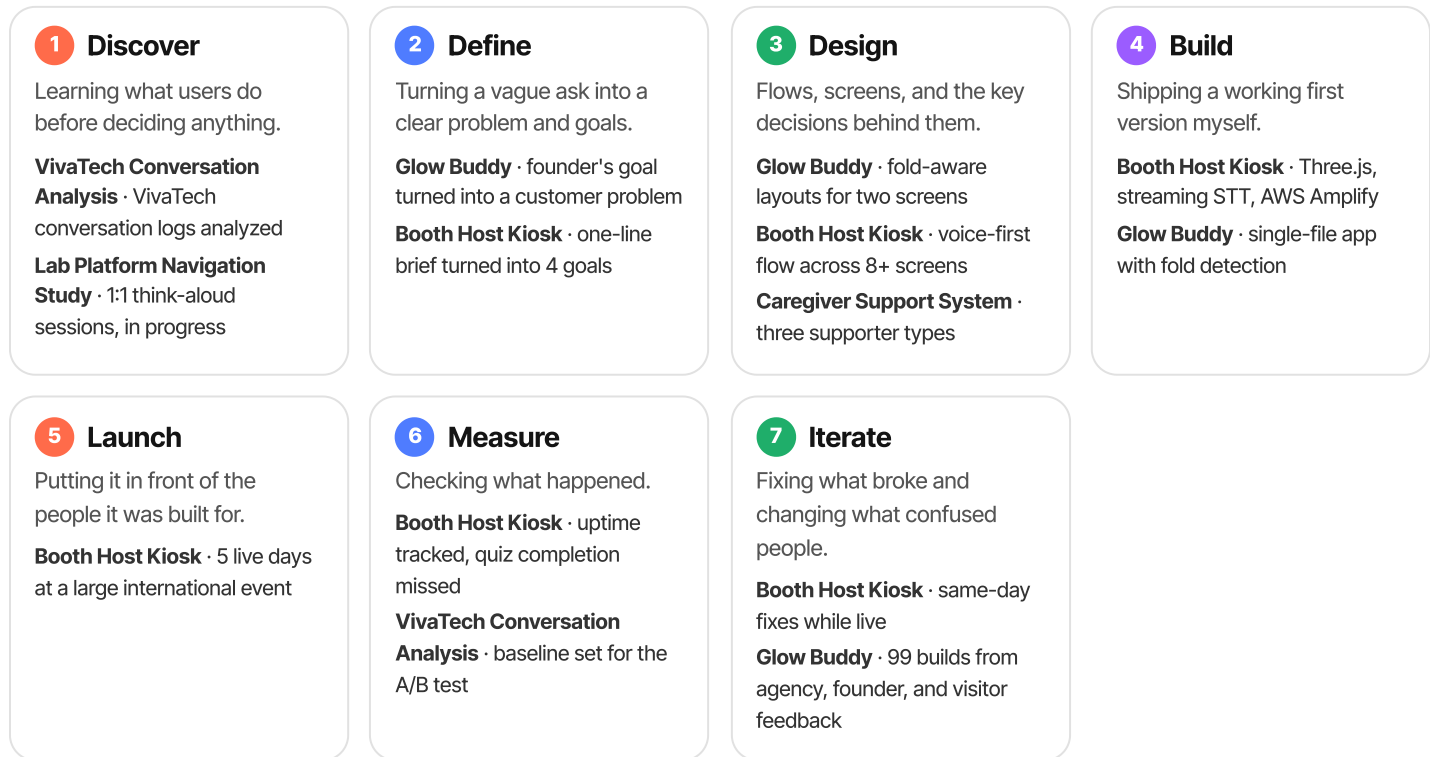
SKILLS ● AND TOOLS ●

Concept design Healthcare Research planning Partnerships Concept design Persona design

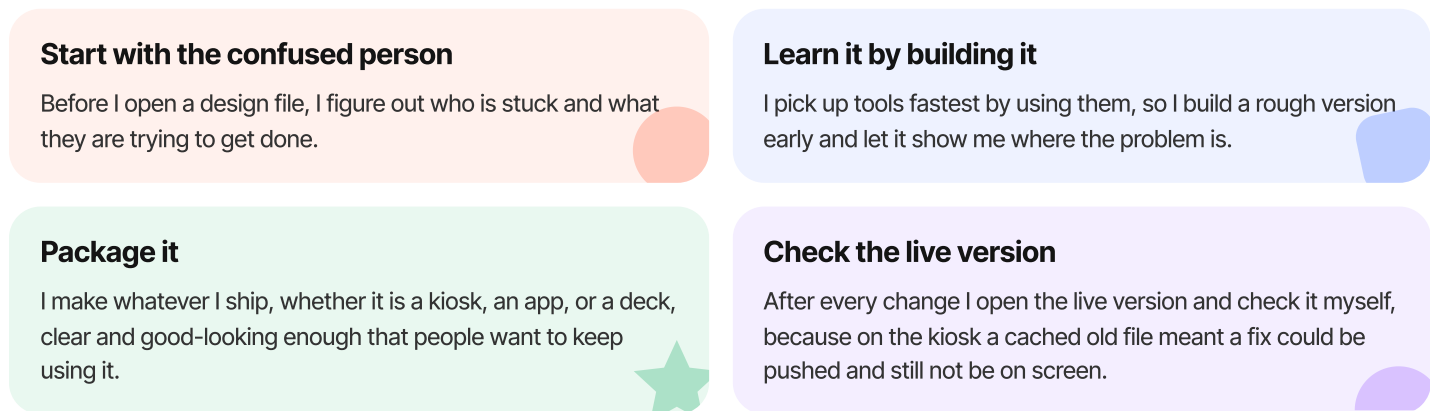
Co-design research Digital humans Healthcare AI

My product loop

Each stage, and which projects show it. The loop runs back to the start, because running the kiosk live showed me what I should measure on the next project.



How I work



"Give him one thing to do, and he comes back having done ten."

Kileon's founder, translated from Korean.

Want the details behind any of these? I am happy to walk through any project, the calls I made, and what I would do differently.

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